

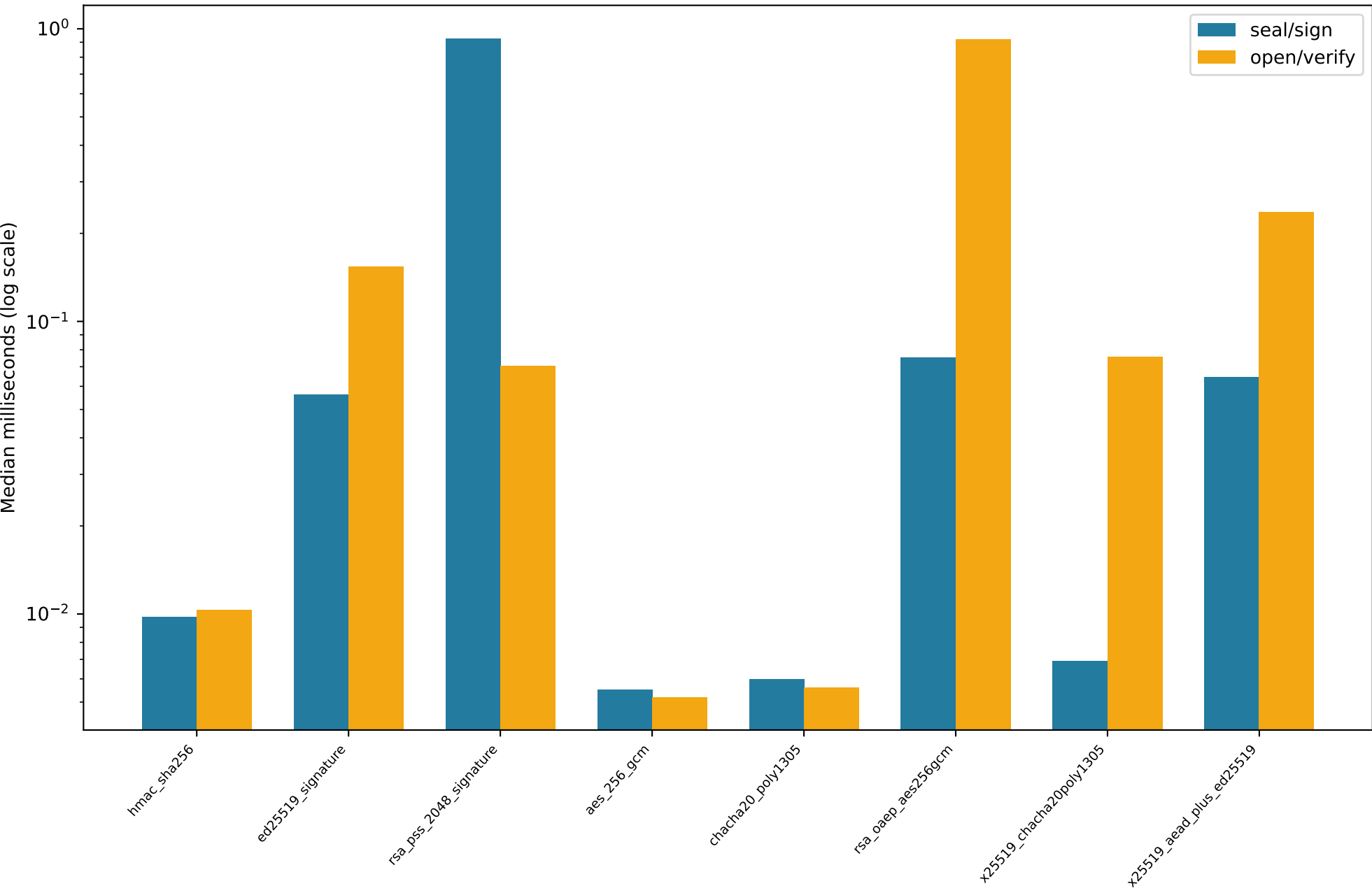
Cryptographic payload alternatives (new)

Actual cryptography-library implementations. HMAC is the current-proposal component reference, not a native timing/overhead measurement; rows with different properties are not drop-in equivalent.

Algorithm	Role	Category	Bytes	Seal μ ms	Seal p95	Open/verify μ	Extra bytes	Tamper reject
hmac_sha256	This Work ref.	symmetric MAC	32	0.0097	0.0153	0.0103	32	True
hmac_sha256	This Work ref.	symmetric MAC	256	0.0099	0.0105	0.0106	32	True
hmac_sha256	This Work ref.	symmetric MAC	1024	0.0106	0.0119	0.0113	32	True
ed25519_signature	Comparison	digital signature	32	0.0564	0.0776	0.1542	64	True
ed25519_signature	Comparison	digital signature	256	0.0574	0.0826	0.1526	64	True
ed25519_signature	Comparison	digital signature	1024	0.0604	0.0784	0.1547	64	True
rsa_pss_2048_signature	Comparison	digital signature	32	0.9283	1.1861	0.0704	256	True
rsa_pss_2048_signature	Comparison	digital signature	256	0.9188	1.0258	0.0703	256	True
rsa_pss_2048_signature	Comparison	digital signature	1024	0.9198	1.2841	0.0698	256	True
aes_256_gcm	Comparison	symmetric AEAD	32	0.0055	0.0074	0.0052	28	True
aes_256_gcm	Comparison	symmetric AEAD	256	0.0054	0.0065	0.0051	28	True
aes_256_gcm	Comparison	symmetric AEAD	1024	0.0062	0.0071	0.0057	28	True
chacha20_poly1305	Comparison	symmetric AEAD	32	0.0060	0.0070	0.0056	28	True
chacha20_poly1305	Comparison	symmetric AEAD	256	0.0061	0.0073	0.0057	28	True
chacha20_poly1305	Comparison	symmetric AEAD	1024	0.0068	0.0118	0.0064	28	True
rsa_oaep_aes256gcm	Comparison	public-key hybrid encrypt	32	0.0751	0.1841	0.9234	284	True
rsa_oaep_aes256gcm	Comparison	public-key hybrid encrypt	256	0.0760	0.1225	0.9304	284	True
rsa_oaep_aes256gcm	Comparison	public-key hybrid encrypt	1024	0.0738	0.1221	0.9196	284	True
x25519_chacha20poly1305	Comparison	authenticated public-key	32	0.0069	0.0076	0.0756	28	True
x25519_chacha20poly1305	Comparison	authenticated public-key	256	0.0071	0.0080	0.0754	28	True
x25519_chacha20poly1305	Comparison	authenticated public-key	1024	0.0078	0.0100	0.0762	28	True
x25519_aead_plus_ed25519	Comparison	encrypt-then-sign compos	32	0.0647	0.0826	0.2364	92	True
x25519_aead_plus_ed25519	Comparison	encrypt-then-sign compos	256	0.0672	0.1058	0.2394	92	True
x25519_aead_plus_ed25519	Comparison	encrypt-then-sign compos	1024	0.0702	0.0785	0.2381	92	True

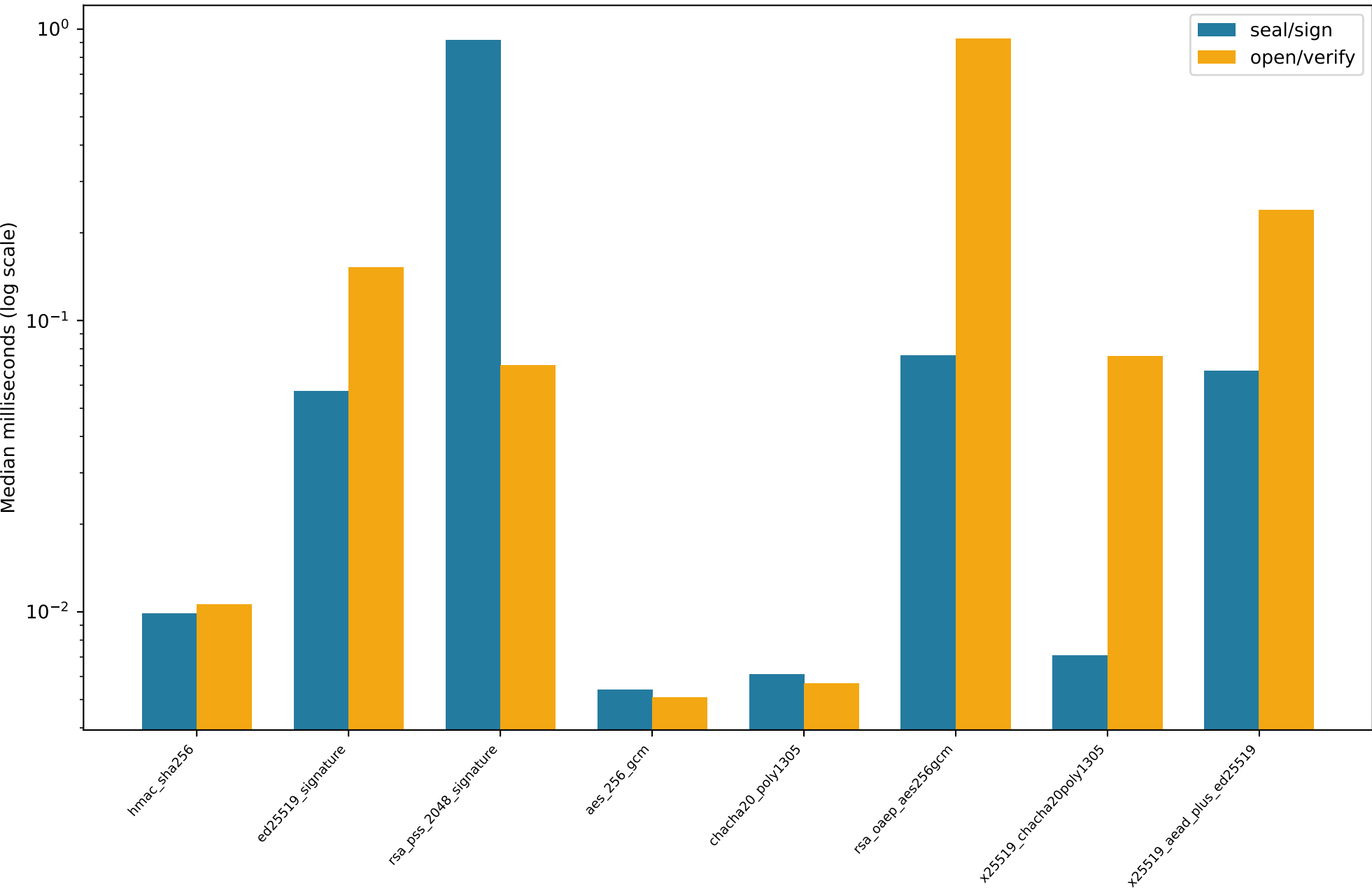
Seal and open/verify timing — 32 byte payload

Measured on this host with generated one-run keys and real operations; not literature values.



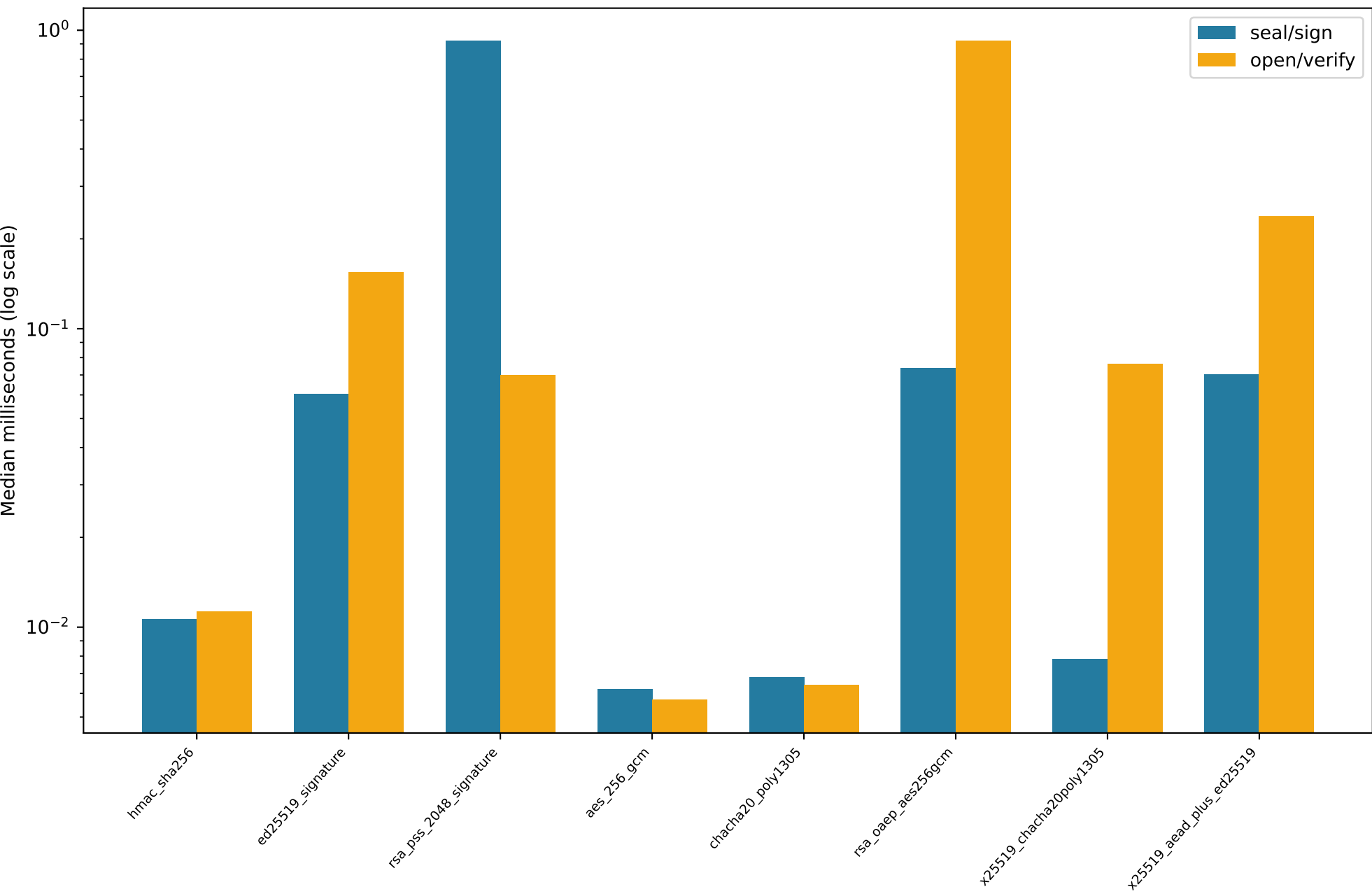
Seal and open/verify timing — 256 byte payload

Measured on this host with generated one-run keys and real operations; not literature values.



Seal and open/verify timing — 1024 byte payload

Measured on this host with generated one-run keys and real operations; not literature values.



Security properties and non-equivalence

True means provided under the listed key-management assumptions; this table is not a certification or formal security proof.

Algorithm	Confidentiality	Integrity	Shared-key / peer auth	Public verify	Assumptions / scope
hmac_sha256	False	True	True	False	Reference primitive measured with full 32
ed25519_signature	False	True	False	True	RFC 8032 Ed25519 detached signature; s
rsa_pss_2048_signature	False	True	False	True	RSA-2048 PSS/SHA-256; payload remains
aes_256_gcm	True	True	True	False	AES-256-GCM shared-key AEAD.
chacha20_poly1305	True	True	True	False	ChaCha20-Poly1305 shared-key AEAD.
rsa_oaep_aes256gcm	True	True	False	False	RSA-OAEP/SHA-256 wraps a fresh AES-25
x25519_chacha20poly1305	True	True	True	False	X25519 static-DH + HKDF-SHA-256 + Ch
x25519_aead_plus_ed25519	True	True	True	True	Explicit X25519 AEAD + detached Ed255

This Work / proposal baseline

This page distinguishes the current native video proposal from primitive-only comparison rows.

Layer	Current proposal and benchmark scope
Video carrier	Modify H.264 CAVLC coefficients in the native compressed-domain path; media benchmark
Authentication	Native H.264 CAVLC stream modification with per-frame HMAC-SHA-256 truncated to 16
Crypto table HMAC row	Primitive reference only: Python cryptography full 32-byte HMAC output. Native frame-ta
ZKP relation	Groth16/PLONK tests prove a payload commitment statement separately; the current vici

Signcryption scope

No standardized signcryption primitive is available in the installed crypto stack. The real measured X25519+AEAD and Ed25519+AEAD composition is labeled precisely and must not be reported as formal signcryption.

Item	Benchmark treatment
X25519 + HKDF + ChaCha20-Poly1305	Actually executed authenticated public-key encryption with pre-authenticated static sender
X25519 AEAD + Ed25519 detached signature	Actually executed encrypt-then-sign composition; publicly verifiable, higher overhead, but
HMAC-SHA-256	Current native CAVLC stream authenticator; integrity/origin among key holders only; no e
AES-GCM / ChaCha20-Poly1305	Shared-key confidentiality and integrity; identity depends on key distribution.
RSA-PSS / Ed25519	Publicly verifiable signatures, but payload is not confidential.
RSA-OAEP + AES-GCM	Public-key recipient confidentiality; does not authenticate the sender.